QUEEN'S UNIVERSITY MIDTERM VERSION A
FACULTY OF ARTS AND SCIENCE
SCHOOL OF COMPUTING

CISC 220 001 – Professor Burton Ma
October 24, 9:30AM

INSTRUCTIONS TO STUDENTS:
This midterm test is 50 MINUTES in length.
There are 5 questions worth a total of 30 marks in this examination.
Please answer all questions in the question booklet.
No aids are allowed.

Put your name and student number in the space indicated below.

GOOD LUCK!

PLEASE NOTE:
Asking questions during the test is discouraged.
Do your best to answer exam questions as written.

This material is copyrighted and is for the sole use of students registered in CISC 220 and writing this quiz. This material shall not be distributed or disseminated. Failure to abide by these conditions is a breach of copyright and may also constitute a breach of academic integrity under the University Senate's Academic Integrity Policy Statement.

NAME:

STUDENT NUMBER:

Question 1 (1 mark each, 5 marks total):

Consider a filesystem having the directory structure shown below:

```
/
└── home
    ├── user1
    │   └── courses
    │       ├── CISC204
    │       └── CISC220
    │       └── more directories not shown
    └── user2
        └── courses
            ├── CISC204
            └── CISC352
```

Your answer for each of the following parts should be very brief.

A) What would the command `pwd` print when run in the directory `CISC352`?

B) Suppose that the current working directory is `user1`. What command would you use to remove the file named `A1.pdf` from the directory `CISC220`?

C) Suppose that the current working directory is `user1`. What single command would you use to list the non-hidden files in all the directories in the `courses` directory belonging to `user1`?

D) Suppose that the current working directory is `CISC220`. What single command would you use to change to the directory `CISC204` belonging to `user1` using a relative path?

E) Suppose that the current user is one of `user1` or `user2` (but for the purposes of the question, you don't know which) and that they are in their `CISC204` directory. What command could they use to copy all the files in their `CISC204` directory into their home directory?

Question 2 (1 mark each, 7 marks total):

Your answer for each of the following parts should be very brief.

A) Explain what filenames are matched by the expansion `[abc]?.sh`

B) What kind of expansion is used when you want to store the output of a command in a variable? If you don't remember the name of the expansion, then provide an example instead.

C) What is a tee used for?

D) Suppose that the variable `f` stores the name of a file. Complete the following `if` statement so that the string "`directory`" is printed if `f` is the name of a directory.

```
if _____________________________________
    echo "directory"
fi
```

E) Suppose that you have an array variable `a`. Show how to print the length of the array stored in `a`.

F) Provide a regular expression that matches all strings that start and end with an uppercase `A`.

G) Complete the missing parts of the following Bash loop so it outputs the letters `a`, `b`, and `c` on separate lines:

```
_____________________ a b c __________

fi
```

Question 3 (2 marks):

The `tabdate` command outputs the current time in the following format:

```
Wed    Oct    23    12:54:59    EDT    2024
```

where each field of the output is separated from the next using a tab character.

Write a single pipeline that:

- saves the month, day, and current time (2nd, 3rd, and 4th fields) to a file named `now.txt`, and
- prints the current time to standard output

Question 4 (4 marks):

Write a complete Bash script that requires two input arguments where the first argument is a new directory name `d`, and the second argument is the name of an existing, readable, regular file `f` located in the same directory as the script.

The script creates the new directory `d` in the same directory as the script and moves the file `f` into the newly created directory. It then changes the permissions of the file `f` by removing all permissions for all users who are not the owner of the file (you can assume that the owner is the current user).

No error checking is required in this question. You can assume that two input arguments are always given and that all required permissions are given.

Question 5 (12 marks):

Consider a plain text file containing the names of students in a course, one name per line in no particular order. An example of the contents of such a file is:

```
Simpson, Bart
Simpson, Lisa
Wiggum, Ralph
```

Note that the names are always formatted as `lastname, firstname`.

The Bash command will read each line of a file named `fname` into an array `arr`:

```
readarray -t arr < fname
```

For example, if the sample file above is named `students.txt` then

```
readarray -t arr < students.txt
```

creates an array of size 3 where the elements are `"Simpson, Bart"`, `"Simpson, Lisa"`, and `"Wiggum, Ralph"`.

Write a script that performs the following steps:

1. If the number of command line arguments is less than 1, then the script outputs an error message to *standard error* and exits with a status of 1.

2. If one or more command line arguments are given, then the first argument is interpreted as the name of a file containing a list of student names as described above (and any remaining arguments are ignored). Your script outputs the name of each student using their first name, followed by a space, followed by their last name. For example, for the file shown above your script should output (there is no space before the first name):

   ```
   Bart Simpson
   Lisa Simpson
   Ralph Wiggum
   ```

You may assume that if a filename is provided to the script as an argument, then the file exists and is readable. Furthermore, every student has exactly one first name and one last name.

Your script must implement and use the functions shown on the next page. It is up to you to decide how your functions receive their arguments and return their values.

Recall that if `s` is a variable then `${s:offset:len}` expands to the substring of `s` starting at index `offset` and having length `len`.

```
function index_of_comma {
    # finds the index of the first comma in a specified string









}

function get_firstname {
    # gets the first name of a student given their full name
    #   e.g., “Simpson, Bart” has the first name “Bart”









}

function get_lastname {
    # gets the last name of a student given their full name
    #   e.g., “Simpson, Bart” has the last name “Simpson”









}
# the rest of your script starts here
```

EXTRA PAGE

EXTRA PAGE

EXTRA PAGE